

状态压缩动态规划

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例1、pku2288Islands and Bridges

- **Time Limit:** 4000MS**Memory Limit:** 65536K**Total Submissions:** 3371**Accepted:** 806**Description**
- Given a map of islands and bridges that connect these islands, a Hamilton path, as we all know, is a path along the bridges such that it visits each island exactly once. On our map, there is also a positive integer value associated with each island. We call a Hamilton path the best triangular Hamilton path if it maximizes the value described below.

Suppose there are n islands. The value of a Hamilton path $C_1C_2...C_n$ is calculated as the sum of three parts. Let V_i be the value for the island C_i . As the first part, we sum over all the V_i values for each island in the path. For the second part, for each edge C_iC_{i+1} in the path, we add the product $V_i * V_{i+1}$. And for the third part, whenever three consecutive islands $C_iC_{i+1}C_{i+2}$ in the path forms a triangle in the map, i.e. there is a bridge between C_i and C_{i+2} , we add the product $V_i * V_{i+1} * V_{i+2}$.

Most likely but not necessarily, the best triangular Hamilton path you are going to find contains many triangles. It is quite possible that there might be more than one best triangular Hamilton paths; your second task is to find the number of such paths.

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- **Input**
- The input file starts with a number q ($q \leq 20$) on the first line, which is the number of test cases. Each test case starts with a line with two integers n and m , which are the number of islands and the number of bridges in the map, respectively. The next line contains n positive integers, the i -th number being the V_i value of island i . Each value is no more than 100. The following m lines are in the form $x \ y$, which indicates there is a (two way) bridge between island x and island y . Islands are numbered from 1 to n . You may assume there will be no more than 13 islands.
- **Output**
- For each test case, output a line with two numbers, separated by a space. The first number is the maximum value of a best triangular Hamilton path; the second number should be the number of different best triangular Hamilton paths. If the test case does not contain a Hamilton path, the output must be `0 0'.

Note: A path may be written down in the reversed order. We still think it is the same path.

- **Sample Input**
- 2 3 3 2 2 2 1 2 2 3 3 1 4 6 1 2 3 4 1 2 1 3 1 4 2 3 2 4 3 4 **Sample Output**
- 22 3 69 1 **Source**
- [Shanghai 2004](#)

分析

- 题目要求找一条最优的汉密尔顿路
- 设 V_i 为岛 C_i 的价值，汉密尔顿路 $C_1C_2\dots C_n$ 的价值分为3部分：
- 1、各个岛屿的价值之和，即 $\sum_{i=1}^n v_i$
- 2、对于路径中每条边 C_iC_{i+1} ，把 V_i*V_{i+1} 加进总价值中
- 3、路径中3个连续的岛 $C_iC_{i+1}C_{i+2}$ ，如果 C_i 与 C_{i+2} 之间有一条边，那么 $V_i*V_{i+1}*V_{i+2}$ 要加进总价值中

方法一

- 搜索
- 1、 $N*N!$
- 时间复杂度 $O(N!)$, $13!=6227020800$, 超时

方法二

- 状态压缩DP
- 用 $F[i,j,s]$ 表示以 i, j 两个岛屿开头的岛屿经过情况为 s 的最大价值
- s 的表示:已经经过的岛屿记为0, 没有经过的记为1, 得到一个01组成二进制, 把它转为十进制。
- 通过考虑紧跟在 i, j 后面的岛屿 k 得到以下状态转移方程:
- $F[i,j,s] = \max(V_k * V_j + V_i * V_j * V_k \text{ (如果 } a[i,k]=1) + F[j,k,s'] \text{), 满足 } a[j,k]=1$
- 其中 k 满足 $(a[j,k]=1) \text{ and } (s \text{ and } (1 \text{ shl}(k-1))) \neq 0, s' = s \text{ xor } (1 \text{ shl}(k-1))$
- 时间复杂度为 $O(N^3 * 2^N)$, 最坏17997824

例2铺砖问题

- 题目描述**ZJU1100**
- 用 1×2 的 砖头铺满 $N \times M$ 的区域，不能有重叠，一共有多少种方案？
- $1 \leq N, M \leq 11$

例3、炮兵阵地

- 给定一个 $N \times M$ 的炮兵阵地，有些地方为平地标记为P,有些地方为山地标记为L,现在要在
这里布置一些炮兵，炮兵只能放置在平地
处，同时具有下图所示的攻击范围，现在
要布置尽可能多的炮兵，但他们之间不能
互相攻击。
- $N \leq 100, M \leq 8$

